LEARNER GUIDE

Which is larger?

Place Value · Lesson 6. A short guide to walk beside you — read as much or as little as helps. *Connection before curriculum. Always.*

What this lesson is about

To compare two numbers, line them up on the decimal point and compare the columns from the left. The first column where the digits differ decides, and an empty column counts as 0. 0.5 is larger than 0.45: they first differ in the tenths, and 5 tenths beats 4 tenths. Having more digits does not make a number larger.

What you are learning to notice

- Line the numbers up on the point, not on the left or right edge.
- Compare from the left; stop at the first column that differs.
- An empty column counts as 0: 0.3 and 0.300 are equal.
- A longer decimal can be smaller (0.45 < 0.5) or larger (0.55 > 0.5): the columns decide, not the length.
- Each column is worth more than all the columns to its right put together.

Moving through it, one step at a time

Write the numbers one above the other with the decimal points lined up. Starting from the left, compare the digits in each column, treating an empty column as 0. At the first column where they differ, the larger digit belongs to the larger number. To order a set, find the smallest this way, then the next, and so on.

Worked examples

Two full examples to follow. Read them slowly — the aim is to see how each line follows from the one before, not to memorise a rule. It can help to cover the steps and predict the next line before you reveal it.

Example 1 — 0.5 or 0.45?

Which is larger, 0.5 or 0.45?

0.5

Ones		Tenths	Hundredths
0	•	5	

0.45

Ones		Tenths	Hundredths
0	•	4	5

Lined up on the point. The ones match (0 and 0); the tenths differ: 5 against 4.

1. Ones: 0 and 0.	the same
2. Tenths: 5 and 4.	different: 5 tenths is more
3. The 5 hundredths in 0.45 cannot make up a whole tenth.	0.5 is larger

0.5 > 0.45. The longer decimal is the smaller one here.

Example 2 — 2,099 or 2,109?

Which is larger, 2,099 or 2,109?

2,099

Thousands	Hundreds	Tens	Ones
2	0	9	9

2,109

Thousands	Hundreds	Tens	Ones
2	1	0	9

The thousands match; the hundreds decide: 1 hundred beats 0 hundreds.

1. Thousands: 2 and 2.	the same
2. Hundreds: 0 and 1.	different: 1 hundred is more
3. The 99 in 2,099 is less than one hundred.	2,109 is larger

2,109 > 2,099, even though 2,099 has bigger digits at the end.

A gentle reminder

You can pause. You can return. You do not need to complete everything in one sitting.

If you get stuck

Getting stuck is part of doing mathematics, not a sign that something is wrong. There is support built into the lesson, and using it is a strong move:

- **Reconnection Routes** — if an earlier idea feels shaky (tenths, hundredths and thousandths (Lesson 4)), open a short recap and return whenever you are ready.
- **Socratic prompt** — a question to nudge your thinking.
- **Hint** — a little more help with the step in front of you.
- **Check** — try an answer and see how you are doing; a wrong try is useful information.
- **Worked solution** — a full model to compare your thinking against.

How the worked solution unlocks

The worked solution unlocks after you use the Socratic prompt, use the hint, and check two answers. Those steps are the learning — the worked solution is there to confirm your thinking, not to replace it.

Using support is part of learning. It is not cheating.

Recording your thinking

You can show your working in whatever way suits you today — the on-screen Scratchpad, paper, a spoken explanation, or a photo of a sketch. There is no single right format. If you do not finish everything in one sitting, that is completely fine: what you have done is information for your next step, never a failure.

Before you begin

Settle into a comfortable pace and take the steps one at a time. If your attention drifts, pause and come back — the lesson will keep your place. When you are ready, begin.

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